CHAPTER 11 · DERIVATIONS

THERMODYNAMICS OF GASES

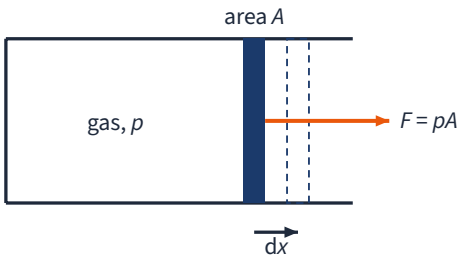
7 derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 WORK DONE BY A GAS

SYLLABUS 11.2(A)

RESULT $W = \int p\,dV$

1. Gas at pressure p pushes a piston of area A with force F = pA.
2. The piston moves out a small distance dx: dW = F dx = pA dx = p dV.
3. From V_1 to V_2: **$W = \int p\,dV$** (from V_1 to V_2) = area under the p–V graph.



2 WORK DONE AT CONSTANT PRESSURE

SYLLABUS 11.2(B)

RESULT $W = p\Delta V = nR\Delta T$

1. Constant p: $W = \int p\,dV = p(V_2 - V_1) =$ **$p\Delta V$**.
2. $pV = nRT$ with p constant: $p\Delta V =$ **$nR\Delta T$**.

3 WORK FROM A P–V GRAPH

SYLLABUS 11.2(C)

RESULT W = area under the p–V curve

1. $W = \int p\,dV$ is the area between the curve and the V-axis from V_1 to V_2.
2. Expansion: W positive (work done by the gas). Compression: W negative.
3. For a closed cycle, the net work = area enclosed by the loop.

4 ΔU FROM A CONSTANT-VOLUME PROCESS

SYLLABUS 11.3(D)

RESULT $\Delta U = nC_{V,m}\Delta T$

1. Constant volume: $\Delta V = 0$, so $W = 0$.
2. First law $Q = \Delta U + W$ gives $Q = \Delta U$.
3. Heat at constant volume $Q = nC_{V,m}\Delta T$, so **$\Delta U = nC_{V,m}\Delta T$**.
4. U of an ideal gas depends only on T, so this holds for any process with the same ΔT.

5 $C_{P,M} - C_{V,M} = R$

SYLLABUS 11.3(E)

RESULT $C_{p,m} - C_{V,m} = R$

1. Heat n moles at constant pressure through ΔT: $Q = nC_{p,m}\Delta T$.
2. $\Delta U = nC_{V,m}\Delta T$ (depends only on ΔT); $W = p\Delta V = nR\Delta T$.
3. First law: $nC_{p,m}\Delta T = nC_{V,m}\Delta T + nR\Delta T$.
4. **$C_{p,m} - C_{V,m} = R$.**

6 MOLAR HEAT CAPACITIES FROM DEGREES OF FREEDOM

SYLLABUS 11.3(F)

RESULT $C_{V,m} = \frac{f}{2}R$, $C_{p,m} = \frac{f+2}{2}R$

1. $U = \frac{f}{2}nRT$, so $\Delta U = \frac{f}{2}nR\Delta T$.
2. Also $\Delta U = nC_{V,m}\Delta T$, so **$C_{V,m} = \frac{f}{2}R$**.
3. $C_{p,m} = C_{V,m} + R = \mathbf{\frac{f+2}{2}R}$.
4. $\gamma = \frac{C_{p,m}}{C_{V,m}} = \frac{f+2}{f}$: monatomic 1.67, diatomic 1.40.

7 WORK DONE IN THE FOUR PROCESSES

SYLLABUS 11.4(F)

RESULT $W = 0;\ p\Delta V;\ nRT\ln\frac{V_2}{V_1};\ \frac{p_1V_1 - p_2V_2}{\gamma - 1}$

1. **Constant volume (isochoric)**: $dV = 0$, so $W = 0$.
2. **Constant pressure (isobaric)**: $W = p\Delta V = nR\Delta T$.
3. **Isothermal**: $p = \frac{nRT}{V}$, so $W = \int \frac{nRT}{V}\,dV = \mathbf{nRT\ln\frac{V_2}{V_1}} = nRT\ln\frac{p_1}{p_2}$.
4. **Adiabatic**: pV^γ = constant, so $pV^\gamma = p_1V_1^\gamma = p_2V_2^\gamma$ and $p = p_1V_1^\gamma V^{-\gamma}$.
5. $W = \int p\,dV = p_1V_1^\gamma \int V^{-\gamma}\,dV = \frac{p_1V_1^\gamma V_2^{1-\gamma} - p_1V_1^\gamma V_1^{1-\gamma}}{1-\gamma}$.
6. Using $p_1V_1^\gamma = p_2V_2^\gamma$: $p_1V_1^\gamma V_2^{1-\gamma} = p_2V_2$ and $p_1V_1^\gamma V_1^{1-\gamma} = p_1V_1$, so $W = \mathbf{\frac{p_1V_1 - p_2V_2}{\gamma - 1}}$ ($= -\Delta U$, because $Q = 0$).

